

VL 405: Design for Testability

Course Code: VL 405

Course Name: Design for Testability

L-T-P-C: 3-0-2-4

Prerequisites: Digital System Design (EE 210) or equivalent

Intended for: Discipline Core for BTech in VLSI & Microelectronics Engineering Students; Discipline Elective for B.Tech in Electrical Engineering students; Free Elective for other B.Tech students

Approval: 61st BoA

Preamble

In the modern era of complex systems and rapid product development, the ability to test effectively and efficiently has become a cornerstone of engineering excellence. Testability is not an afterthought—it is a deliberate design philosophy that ensures systems can be validated, faults can be diagnosed, and quality can be guaranteed throughout the product lifecycle. This course on Design for Testability introduces participants to the principles, strategies, and practices that make systems easier to test, maintain, and evolve. By embedding testability into the design process, engineers reduce costs, accelerate debugging, and improve reliability.

Course topics

- **1. Introduction to design for testability:** Testing during VLSI life cycle, challenges, yield and reject rate. (1 hour)
- **2. Overview of fault models:** Stuck at faults, transistor faults, open and short faults, delay faults and crosstalk, pattern sensitivity and coupling faults. (3 hours)
- **3. Testability measures:** SCOAP testability analysis, controllability and observability, probability based testability analysis, simulation based testability analysis, RTL testability analysis. (6 hours)
- **4. Adhoc testing Architectures:** Design techniques, scan design, test point insertion. (2 hours)
- **5. Logic built in self test:** BIST design rules, test pattern generation, output response analysis, signature analysis, logic BIST architectures, fault coverage enhancement. (5 hours)
- **6. Random access scan design:** random access scan (RAS) flip flop, PRAS design, STAR design, RAS applications. (2 hours)
- **7. Logic Simulation:** logic models, logic model implementation, hardware description languages, true value simulations, delay modeling. (2 hours)
- **8. Test Compression:** test stimulus compression, code based schemes, linear decompression based schemes, broadcast scan based schemes. (2 hours)
- **9. Test-benches:** objectives, stimulus and response, assertion and report generation (2 hours)
- **10. Fault Simulation:** serial fault simulation, parallel fault simulation, deductive fault simulation, parallel pattern fault simulation. (2 hours)
- **11. Linear Feedback Shift Register and its applications on test architectures:** design and analysis of linear feedback shift registers, test pattern generation using linear feedback shift registers, output response capture. (5 hours)

- **12. Memory testing:** SRAM and DRAM fault models, reduced functional model, address decoder faults, coupling faults, March test elements. (2 hours)
- **13. Test vector generation:** notion of ATPG, structural versus functional ATPG, random test generation, D algorithm (2 hours)
- **14. Boundary Scan testing:** digital boundary scan, boundary scan for advanced networks, test access port, bus protocols. (2 hours)
- **15. Analog and mixed signal testing:** Issues with analog testing, difference from digital testing, test parameters, analog fault models, analog test bus. (2 hours)
- **16. Testing technologies and recent trends:** Coping with physical failures, soft errors and reliability issues, FPGA testing, MEMS testing, high speed I/O testing, RF testing. (2 hours)

Laboratory work

Laboratory assignments covering the fundamentals in testing techniques and design of testable architectures. Typical experiments include but not limited to fault model generation and fault detection of combinational and sequential digital logic circuits of varying complexity, verification of testing protocols and standards, etc. (28 hours)

Text Books

1. *VLSI Test Principles and Architectures: Design for Testability* (2024) by Laung-terng Wang, Cheng-Wen Wu, and Xiaoqing Wen, Morgan Kaufmann

Reference Books

1. *Digital Systems Testing and Testable Design* (1994) by Miron Abramovici, Melvin A. Breuer, Wiley.
2. *Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits* (2013) by Michael Bushnell and Vishwani D. Agrawal, Springer.